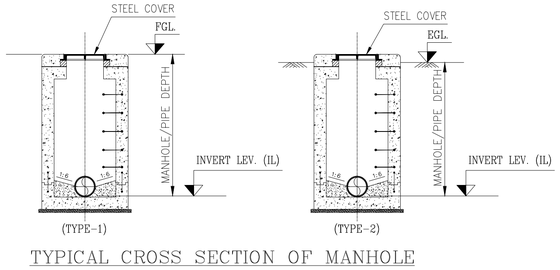
SCHEDULE OF SEWERAGE NETWORK MANHOLES :-

SL. NO.	SEWERAGE MANHOLE	TOP OF MANHOLE COVER LEVEL (M)	ENTRY INVERT LEVEL (M)	EXIT INVERT LEVEL (M)	CONSIDERING EGL. OR FGL. (M)	PIPE (EXIT) DEPTH FROM EGL. OR FGL. (M)	MANHOLE INTERNAL PLAN DIMENSIONS (MM)	CLEAR COVER OPENING DIMENSIONS (MM)	TYPE OF SOLID COVER	CO-ORDINATES AT CENTER OF MANHOLE		TYPE OF MANHOLE
										E (M)	N (M)	
1	MH- 01	912.6	-	911.665	912.6	0.935	Ø900	Ø750	Group-4	E=264220.587	N=2506133.217	TYPE-1
2	MH- 02	912.6	911.515	911.49	912.6	1.11	Ø900	Ø750	Group-4	E=264233.930	N=2506126.585	TYPE-1
3	MH- 03	912.6	911.22	911.195	912.6	1.405	Ø900	Ø750	Group-4	E=264257.251	N=2506141.158	TYPE-1
4	MH- 04	912.6	910.895	910.87	912.6	1.73	Ø900	Ø750	Group-4	E=264282.693	N=2506157.055	TYPE-1
5	MH- 05	912.6	910.665	910.64	912.6	1.96	Ø900	Ø750	Group-4	E=264286.593	N=2506177.095	TYPE-1
6	MH- 06	913	910.42	910.395	913	2.605	Ø1500	Ø750	Group-4	E=264274.426	N=2506195.434	TYPE-1
7	MH- 07	913	910.095	910.07	913	2.93	Ø1500	Ø750	Group-4	E=264256.970	N=2506219.784	TYPE-1
8	LS- 1	913.15	909.795	912	913	-	-	-	-	-	-	-
9	MH1- 01A	913	911.95	911.925	913	1.075	Ø900	Ø750	Group-4	E=264239.684	N=2506249.900	TYPE-1
10	MH1- 01B	913	911.745	911.72	913	1.28	Ø900	Ø750	Group-4	E=264223.995	N=2506240.097	TYPE-1
11	MH1- 01/2	913	-	912	913	1	Ø900	Ø750	Group-4	E=264222.514	N=2506271.717	TYPE-1
12	MH1- 01/1	913	911.875	911.85	913	1.15	Ø900	Ø750	Group-4	E=264229.032	N=2506261.286	TYPE-1
13	MH1- 01	913	911.575 (From MH1- 01B) 911.695 (From MH1- 01/1)	911.55	913	1.45	Ø900	Ø750	Group-4	E=264216.099	N=2506252.733	TYPE-1
14	MH1- 02	913	911.27	911.245	913	1.755	Ø900	Ø750	Group-4	E=264191.930	N=2506237.630	TYPE-1
15	MH1- 03	913	911.065	911.04	913	1.96	Ø900	Ø750	Group-4	E=264176.496	N=2506227.986	TYPE-1
16	MH1- 04	913	910.905	910.88	913	2.12	Ø1200	Ø750	Group-4	E=264169.183	N=2506239.689	TYPE-1
17	MH1- 05	913	910.585	910.56	912.9	2.34	Ø1200	Ø750	Group-4	E=264143.741	N=2506223.791	TYPE-2
18	MH1- 06A	913	-	912	913	1	Ø900	Ø750	Group-4	E=264159.911	N=2506274.341	TYPE-1
19	MH1- 06B	913	911.8	911.775	913	1.225	Ø900	Ø750	Group-4	E=264142.950	N=2506263.742	TYPE-1
20	MH1- 06	913	911.57	911.545	913	1.455	Ø1500	Ø750	Group-4	E=264125.565	N=2506252.879	TYPE-1
21	MH1- 07A	913.2	910.21	910.185	913.1	2.915	Ø1200	Ø750	Group-4	E=264115.916	N=2506225.860	TYPE-2
22	MH1- 07	913.2	909.930 (From MH1- 07A) 911.295 (From MH1- 06)	909.905	913.1	3.195	Ø1500	Ø750	Group-4	E=264105.159	N=2506243.076	TYPE-2
23	LS-2	913.25	909.685	912.1	913.1	-	-	-	-	-	-	-
24	MH1- 08	913.1	911.97	911.945	913.1	1.155	Ø900	Ø750	Group-4	E=264078.899	N=2506254.908	TYPE-1
25	MH1- 09	913.1	911.875	911.85	913.1	1.25	Ø900	Ø750	Group-4	E=264075.189	N=2506260.844	TYPE-1
26	MH1- 10	913.1	911.55	911.525	913.1	1.575	Ø900	Ø750	Group-4	E=264049.748	N=2506244.947	TYPE-1
27	MH1- 11A	913.1	-	912.1	913.1	1	Ø900	Ø750	Group-4	E=264041.394	N=2506225.576	TYPE-1
28	MH1- 11B	913.1	912.03	912.005	913.1	1.095	Ø900	Ø750	Group-4	E=264034.535	N=2506224.002	TYPE-1
29	MH1- 11	913.1	911.285 (From MH1- 10) 911.905 (From MH1- 11B)	911.26	913.1	1.84	Ø900	Ø750	Group-4	E=264029.236	E=264029.236	TYPE-1
30	MH1- 12	913.1	911.005	910.98	913.1	2.12	Ø1200	Ø750	Group-4	E=264008.459	N=2506219.500	TYPE-1
31	MH1- 13	913.1	910.68	910.655	913.1	2.445	Ø1200	Ø750	Group-4	E=263982.328	N=2506202.818	TYPE-1
32	MH1- 14	913.1	910.355	910.33	913.1	2.77	Ø1500	Ø750	Group-4	E=263956.887	N=2506186.921	TYPE-1
33	MH1- 15	913.1	910.1	910.075	913.1	3.025	Ø1500	Ø750	Group-4	E=263937.382	N=2506174.733	TYPE-1
34	MH3- 01	913.1	-	912.1	913.1	1	Ø900	Ø750	Group-4	E=263945.330	N=2506162.012	TYPE-1
35	LS- 3	913.25	910.000 (From MH1- 15) 912.025 (From MH3- 01)	912.1	913.1	-	-	-	-	-	-	-
36	MH1- 16	914	911.93	911.905	913.9	1.995	Ø900	Ø750	Group-4	E=263916.763	N=2506153.005	TYPE-2
37	MH1- 17	914.35	911.75	911.725	914.25	2.525	Ø1500	Ø750	Group-4	E=263896.833	N=2506140.551	TYPE-2
38	MH1- 18	914.25	911.605	911.58	914.15	2.57	Ø1500	Ø750	Group-4	E=263891.813	N=2506123.174	TYPE-2
39	MH1- 19A	913.1	-	912.1	913.1	1	Ø900	Ø750	Group-4	E=263918.521	N=2506083.262	TYPE-1
40	MH1- 19	913.7	911.380 (From MH1- 18) 911.900 (From MH1- 19A)	911.355	913.6	2.245	Ø1200	Ø750	Group-4	E=263906.385	N=2506099.852	TYPE-2
41	MH1- 20A/3	913.1	-	912.1	913.1	1	Ø900	Ø750	Group-4	E=263923.616	N=2506043.406	TYPE-1
42	MH1- 20A/2	913.45	911.895	911.87	913.1	1.23	Ø900	Ø750	Group-4	E=263906.231	N=2506032.542	TYPE-2
43	MH1- 20A/1	913.55	911.7	911.675	913.1	1.425	Ø900	Ø750	Group-4	E=263899.724	N=2506048.522	TYPE-2
44	MH1- 20A	913.75	911.415	911.39	913.1	1.71	Ø900	Ø750	Group-4	E=263883.001	N=2506068.383	TYPE-2
45	MH1- 20	913.9	911.125 (From MH1- 19) 911.270 (From MH1- 20A)	911.1	913.8	2.7	Ø1500	Ø750	Group-4	E=263879.301	N=2506079.980	TYPE-2
46	MH1- 21	914	910.9	910.875	913.9	3.025	Ø1500	Ø750	Group-4	E=263853.860	N=2506064.083	TYPE-2
47	MH1- 22	914.05	910.725	910.7	913.95	3.25	Ø1500	Ø750	Group-4	E=263835.202	N=2506052.424	TYPE-2
48	MH2- 01	913.1	-	912.075	913.1	1.025	Ø900	Ø750	Group-4	E=263975.338	N=2505976.674	TYPE-1
49	MH2- 02						VOID					
50	MH2- 03	913.3	911.65	911.625	913.2	1.575	Ø900	Ø750	Group-4	E=263945.857	N=2505962.144	TYPE-2
51	MH2- 04	913.3	911.325	911.3	913.2	1.9	Ø1200	Ø750	Group-4	E=263922.112	N=2505947.306	TYPE-1
52	MH2- 05	913.3	911	910.975	913.2	2.225	Ø1200	Ø750	Group-4	E=263896.670	N=2505931.409	TYPE-1
53	MH2- 06	913.3	910.825	910.8	913.2	2.4	Ø1200	Ø750	Group-4	E=263888.296	N=2505925.940	TYPE-2
54	MH2- 07	913.4	910.625 (From MH2- 06) 910.770 (From MH6- 20)	910.6	913.3	2.7	Ø1500	Ø750	Group-4	E=263864.711	N=2505921.226	TYPE-2
55	MH2- 08	913.5	910.4	910.375	913.4	3.025	Ø1500	Ø750	Group-4	E=263854.961	N=2505936.830	TYPE-2
56	MH4- 01	913.55	-	912.31	913.45	1.14	Ø900	Ø750	Group-4	E=263838.285	N=2505943.513	TYPE-2
57	LS- 4	913.6	910.23 (From MH2- 08) 912.22 (From MH4- 01)	912.45	913.45	-	-	-	-	-	-	-

SCHEDULE OF SEWERAGE NETWORK MANHOLES :-

SL. NO.	SEWERAGE MANHOLE	TOP OF MANHOLE COVER LEVEL (M)	ENTRY INVERT LEVEL (M)	EXIT INVERT LEVEL (M)	CONSIDERING EGL. OR FGL. (M)	PIPE (EXIT) DEPTH FROM EGL. OR FGL. (M)	MANHOLE INTERNAL PLAN DIMENSIONS (MM)	CLEAR COVER OPENING DIMENSIONS (MM)	TYPE OF SOLID COVER	CO-ORDINATES AT CENTER OF MANHOLE		TYPE OF MANHOLE
										E (M)	N (M)	
58	MH2- 09	13.55	912.405	912.38	913.45	1.07	Ø900	Ø750	Group-4	E=263843.037	N=2505955.911	TYPE-2
59	MH2- 10	913.7	912.13	912.105	913.6	1.495	Ø900	Ø750	Group-4	E=263827.140	N=2505981.352	TYPE-2
60	MH2- 11	913.75	911.805	911.78	913.65	1.87	Ø900	Ø750	Group-4	E=263811.242	N=2506006.794	TYPE-2
61	MH2- 12	913.85	911.52	911.495	913.75	2.255	Ø1200	Ø750	Group-4	E=263797.464	N=2506028.843	TYPE-2
62	LS- 5	914	911.330 (From MH2- 12) 910.520 (From MH1- 22)	912.85	913.85	-	-	-	-	-	-	-
63	MH1- 23	914	912.715	912.69	913.9	1.21	Ø900	Ø750	Group-4	E=263798.872	N=2506057.728	TYPE-2
64	MH1- 24	914.35	912.5	912.475	914.25	1.775	Ø900	Ø750	Group-4	E=263783.636	N=2506082.109	TYPE-2
65	MH1- 25	914.6	912.365	912.34	914.5	2.16	Ø1200	Ø750	Group-4	E=263774.893	N=2506096.102	TYPE-2
66	MH1- 26	914.95	912.14	912.115	914.85	2.735	Ø1500	Ø750	Group-4	E=263758.995	N=2506121.543	TYPE-2
67	MH1- 27	914.85	911.915	911.89	914.75	2.86	Ø1500	Ø750	Group-4	E=263743.098	N=2506146.985	TYPE-2
68	MH1- 28	914.85	911.75	911.725	914.75	3.025	Ø1500	Ø750	Group-4	E=263731.969	N=2506164.794	TYPE-2
69	MH5- 01	915.05	-	913.9	915.05	1.15	Ø900	Ø750	Group-4	E=263759.747	N=2506236.251	TYPE-1
70	MH5- 02	915.05	913.85	913.825	915.05	1.225	Ø900	Ø750	Group-4	E=263764.264	N=2506238.396	TYPE-1
71	MH5- 03	915.05	913.675	913.65	915.05	1.4	Ø900	Ø750	Group-4	E=263771.292	N=2506225.217	TYPE-1
72	MH5- 04	914.8	913.505	913.48	914.8	1.32	Ø900	Ø750	Group-4	E=263776.176	N=2506212.935	TYPE-1
73	MH5- 05	914.8	913.375	913.35	914.8	1.45	Ø900	Ø750	Group-4	E=263767.271	N=2506207.371	TYPE-1
74	MH5- 06A	914.8	-	913.8	914.8	1	Ø900	Ø750	Group-4	E=263785.236	N=2506178.622	TYPE-1
75	MH5- 06	914.8	913.19 (From MH5- 05) 913.635 (From MH5- 06A)	913.165	914.8	1.635	Ø900	Ø750	Group-4	E=263776.492	N=2506192.615	TYPE-1
76	MH5- 07	915.1	912.875	912.85	915	2.15	Ø1200	Ø750	Group-4	E=263751.983	N=2506177.300	TYPE-2
77	LS- 6	914.9	911.690 (From MH1- 28) 912.665 (From MH5- 07)	913.75	914.75	-	-	-	-	-	-	-
78	MH1- 29	914.85	913.715	913.69	914.75	1.06	Ø900	Ø750	Group-4	E=263731.705	N=2506174.652	TYPE-2
79	MH1- 30	914.85	913.49	913.465	914.75	1.285	Ø900	Ø750	Group-4	E=263715.808	N=2506200.093	TYPE-2
80	MH1- 31	914.9	913.265	913.24	914.8	1.56	Ø900	Ø750	Group-4	E=263699.910	N=2506225.535	TYPE-2
81	MH1- 32A	915.5	914.42	914.395	915.5	1.105	Ø900	Ø750	Group-4	E=263692.493	N=2506256.275	TYPE-1
82	MH1- 32	915.5	913.040 (From MH1- 31) 914.295 (From MH1- 32A)	913.015	915.5	2.485	Ø1200	Ø750	Group-4	E=263684.013	N=2506250.976	TYPE-1
83	MH1- 33A	915.5	-	914.5	915.5	1	Ø900	Ø750	Group-4	E=263701.720	N=2506305.670	TYPE-1
84	MH1- 33B	915.25	914.28	914.255	915.15	0.895	Ø900	Ø750	Group-4	E=263683.063	N=2506294.012	TYPE-2
85	MH1- 33C	915.15	914.085	914.06	915.05	0.99	Ø900	Ø750	Group-4	E=263668.646	N=2506285.004	TYPE-2
86	MH1- 33D	915.5	913.79	913.765	915.5	1.735	Ø900	Ø750	Group-4	E=263682.954	N=2506262.106	TYPE-1
87	MH1- 33	915.5	912.945 (From MH1- 32) 913.715 (From MH1- 33D)	912.92	915.5	2.58	Ø1500	Ø750	Group-4	E=263678.713	N=2506259.457	TYPE-1
88	FSLP	915.65	912.905	914.5	915.5	-	-	-	-	-	-	-



TYPICAL CROSS SECTION OF MANHOLE

LEGENDS: –

LS = LIFTING STATION
FSLP = FINAL SEWERAGE LIFTING PIT
EGL = EXISTING GROUND LEVEL
FGL = FINISHED GROUND LEVEL
SP = SOIL PIT
WP = WASTE PIT

NOTES:~

1. ALL DIMENSIONS ARE IN MILLIMETRES AND LEVELS AND INVERT LEVEL & CO-ORDINATES ARE IN METRES UNLESS NOTED OTHERWISE.
2. INVERT LEVEL (IL) HAVE BEEN INDICATED FOR POINT OF SEWERAGE DRAIN AND UNIFORM SLOPE SHOULD BE PROVIDED BETWEEN THE MANHOLE.
3. MANHOLE LOCATIONS NEAR TOILET ARE TENTATIVE.
4. LENGTH OF SEWER WATER PIPE BETWEEN MANHOLES ARE INDICATED IN THIS DRAWING. THESE LENGTHS ARE APPROXIMATE (HORIZONTAL) AND TO BE ADJUSTED DURING CONSTRUCTION FOR PROPER FIXING.
5. SEWERAGE COLLECTION PIPE/DRAIN SHALL BE PROVIDED FROM TOILET/OTHER FACILITIES LOCATIONS UP TO MANHOLES WHEREVER REQUIRED. OTHERWISE CONNECTION PIPE/DRAIN SHALL BE PROVIDED BETWEEN S.P/W.P. AND NEAREST SEWERAGE MANHOLE.
6. THIS DRAWING IS APPLICABLE ONLY FOR SEWERAGE DRAIN NETWORK AND MANHOLE LOCATIONS. ALL OTHER INFORMATION, eg. LEVEL(EC.),FACILITY OUTLINE,GRADING ETC. ARE SHOWN IN THIS DRAWING FOR INFORMATION ONLY.
7. INVERT LEVELS (IL) HAVE BEEN INDICATED FOR THE MANHOLES AND UNIFORM SLOPE SHOULD BE PROVIDED IN THE SANITARY SEWER LINES BETWEEN THE MANHOLES.
8. PIPE BETWEEN LIFTING STATION (LS) AND NEXT MANHOLE SHALL BE PRESSURE PIPE. FOR LAYOUT AND DETAILS REFER UTILITY DRAWINGS.
9. FOR S.P./W.P. LOCATION AND DETAILS REFER ARCHITECTURAL DRAWINGS.
10. THIS DRAWING SHOULD BE READ IN CONJUNCTION WITH DRG. NO. MD-216-49EE-EG-CV-DED-1001, MD-216-49EE-EG-CV-DED-1002 & MD-216-49EE-EG-CV-DED-1003.

LIST OF DRAWING NO. OF LIFTING STATION :-

LIFTING STATION	DRAWING NO.
LS-1	MD-216-49EE-EG-CV-GAD-1001
LS-2	MD-216-49EE-EG-CV-GAD-1002
LS-3	MD-216-49EE-EG-CV-GAD-1003
LS-4	MD-216-49EE-EG-CV-GAD-1004
LS-5	MD-216-49EE-EG-CV-GAD-1005
LS-6	MD-216-49EE-EG-CV-GAD-1006

DOCUMENT NUMBER	REFERENCE DOCUMENTS	REV	DATE	REVISION DESCRIPTION	DRN	CHK	ENG	APP	EMP
		D01	11/05/2023	AS-BUILT	PML	BJB	APEX	SJM	
		C07	08/02/2022	ISSUED FOR CONSTRUCTION AND REVISED AS MARKED C07	PML	BJB	APEX	SJM	
		C06	22/12/2021	ISSUED FOR CONSTRUCTION AND REVISED AS MARKED C06	PML	BJB	APEX	SJM	
		C05	10/12/2021	ISSUED FOR CONSTRUCTION AND REVISED AS MARKED C05	PML	BJB	APEX	SJM	
		C04	27/08/2021	ISSUED FOR CONSTRUCTION AND REVISED AS MARKED C04	PML	BJB	APEX	SJM	
MD-216-49EE-EG-CV-DED-1003	LAYOUT & DETAILS OF SEWERAGE NETWORK	C03	19/04/2021	ISSUED FOR CONSTRUCTION AND REVISED AS MARKED C03	APEX	BJB	APEX	SJM	
MD-216-49EE-EG-CV-DED-1002	LAYOUT & DETAILS OF SEWERAGE NETWORK	C02	20/01/2021	ISSUED FOR CONSTRUCTION AND REVISED AS MARKED C02	APEX	BJB	APEX	SJM	
MD-216-49EE-EG-CV-DED-1001	LAYOUT & DETAILS OF SEWERAGE NETWORK	C01	23/07/2020	ISSUED FOR CONSTRUCTION AND REVISED AS MARKED C01	APEX	SNM	APEX	SJM	
MD-216-1000-EG-GE-PPD-1002	PROCESS PLANT LAYOUT	A01	23/06/2020	ISSUED FOR REFERENCE	APEX	SNM	APEX	SJM	

RESERVER FOR VENDOR LOGO AND DRAWING NUMBER

LARSEN & TOUBRO DRAWING NUMBER: 019045_910_C_V_24021_004 — REVISION: 8
OUTOTEC DRAWING NUMBER: 900707__GAU01__AG90__CTB17__50003 — DOCUMENT ID: OU100673153 — REVISION: 8

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DRAWN	APEX	19/06/2020	MANSOURAH - MASSARAH GOLD PROJECT
CHECKED	SNM	22/06/2020	DETAIL DRAWING
ENGINEER	APEX	17/06/2020	PROCESS PLANT SEWERAGE NETWORK
APPROVED	SJM	23/06/2020	SCHEDULE OF SEWERAGE NETWORK MANHOLE
EMPLOYER			
SCALE	1:1	A1	DRAWING NUMBER: MD-216-49EE-EG-CV-DED-1004 — REV: D01